

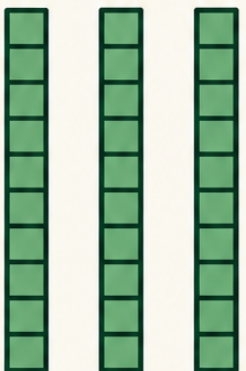


Understanding Tens and Ones

Every digit has a place and a value.

CLASS 13

34

TENS	ONES
 3 tens = 30	 4 ones = 4

$$34 = 30 + 4$$

34 has **3** tens and **4** ones.

10 ones make 1 ten.

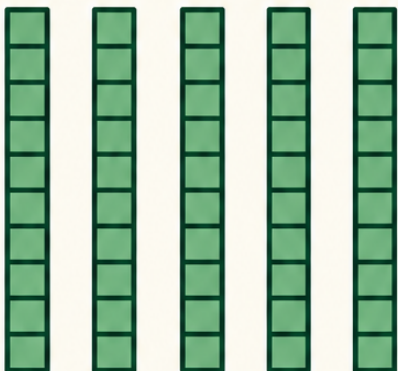


Understanding Tens and Ones

Every digit has a place and a value.

CLASS 13

52

TENS	ONES
 5 tens = 50	 2 ones = 2

$$52 = 50 + 2$$

52 has **5** tens and **2** ones.

10 ones make 1 ten.

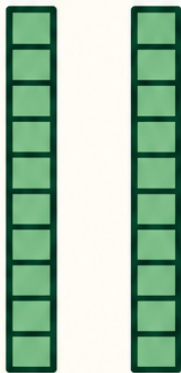


Understanding Tens and Ones

Every digit has a place and a value.

CLASS 13

26

TENS	ONES
 2 tens = 20	 6 ones = 6

$$26 = 20 + 6$$

26 has **2** tens and **6** ones.

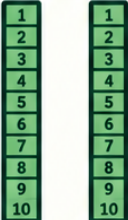
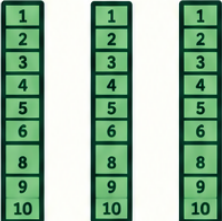
10 ones make 1 ten.



Representing Numbers with Models

Use tens rods and ones squares to add.

CLASS 13

Number	Tens	Ones
23		
+ 14		
37		

2 tens + 1 ten = **3 tens** | 3 ones + 4 ones = **7 ones**

$$23 + 14 = 37$$

Combine tens with tens and ones with ones.



Adding Tens and Ones Separately

Break each number into tens and ones.

CLASS 13

$$32 + 25$$

$$32 = 30 + 2$$

tens ones

$$25 = 20 + 5$$

tens ones

1. Add the tens

$$30 + 20 = 50$$

2. Add the ones

$$2 + 5 = 7$$

3. Combine the parts

$$50 + 7 = 57$$

$$32 + 25 = 57$$

Add tens to tens and ones to ones.



Aligning Numbers Vertically

Write tens under tens and ones under ones.

CLASS 13

	TENS	ONES
	4	2
+	3	5
	7	7

Tens column

$$4 \text{ tens} + 3 \text{ tens} = 7 \text{ tens}$$

Ones column

$$2 \text{ ones} + 5 \text{ ones} = 7 \text{ ones}$$

$$42 + 35 = 77$$

Keep each digit in its correct place.



Aligning Numbers Vertically

Addition with regrouping (carrying).

CLASS 13

	TENS	ONES
Carry 1 ten →	1	
	4	7
+	3	5
	8	2

1. Add the ones
 $7 + 5 = 12$ ones
 Regroup: 12 ones = 1 ten + 2 ones.

2. Add the tens
 $1 + 4 + 3 = 8$ tens
 Include the carried ten.

$$47 + 35 = 82$$

Write 2 in the ones column and carry 1 to the tens.

Comparison only: this poster shows regrouping. Class 13 practice does not require it.



Adding the Ones First

Start with the ones, then add the tens.

CLASS 13

	TENS	ONES
	4	6
+	2	3
	6	9

1.

Add the ones

$$6 + 3 = 9$$

Write 9 in the ones column.

2.

Add the tens

$$4 + 2 = 6$$

Write 6 in the tens column.

$$46 + 23 = 69$$

Begin on the right: ones first, tens next.



When Regrouping Is Not Needed

If the ones total less than 10, no new ten is formed.

CLASS 13

	TENS	ONES
	5	2
+	1	6
	6	8

1.

Check the ones

$$2 + 6 = 8$$

8 is less than 10: keep 8 ones.

2.

Add the tens

$$5 + 1 = 6$$

5 tens + 1 ten = 6 tens.

$$52 + 16 = 68$$

No regrouping is needed in this example.



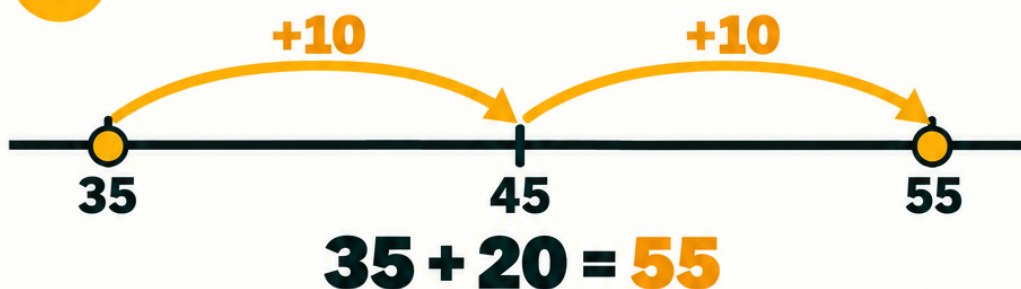
Adding on a Number Line

Start at 35 and move right to add 23.

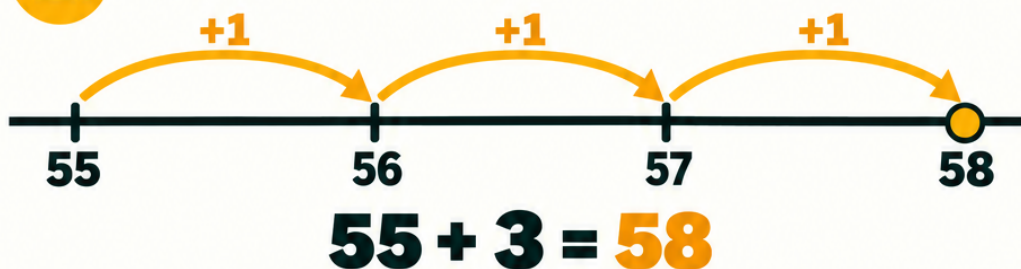
CLASS 13

$$35 + 23 = 58$$

1. Add the tens: +20



2. Add the ones: +3



The second number line zooms in to show ones.

$$23 = 20 + 3$$
$$35 + 23 = 58$$

Move right when adding positive numbers.



Solving Addition Word Problems

Read, choose an operation, and solve.

CLASS 13

A class has 24 pencils. The teacher brings 13 more pencils. How many pencils are there altogether?



1. Find the information

Pencils in the class: **24**

More pencils: **13**

2. Add the amounts

$$24 + 13$$

$$\text{Ones: } 4 + 3 = 7$$

$$\text{Tens: } 2 + 1 = 3$$

3. Answer the question

$$24 + 13 = 37$$

There are **37** pencils altogether.

Check: $20 + 10 = 30$ and $4 + 3 = 7$.

Include the unit in your answer: **pencils**.